pfi

Python for Fixed Income

COURSE 5B · TIME SERIES MODELS

Course 5B Capstone

WEEK 2 · DAY 5

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The brief

The desk's question, in the past tense

Two daily series in bp: the 2s10s slope (real) and an invented HY spread. Fitted on the development years, 2 Jan 2015 to 30 Dec 2022, did any model score closer to each day's value over the held-out years, 3 Jan 2023 to 2 Oct 2026, than the last value carried forward? And did a volatility model beat a constant for the size of each change?

The workflow

1. Repair, with a change log
2. Stationarity and memory
3. Baselines and models, walk-forward
4. GARCH for the size of a change
5. Choose on development only
6. Open the held-out years once

If nothing beats the last value, that is the result, and you say so.

Names for a model's output

The words for what a model gives, in this capstone as on every day:

Allowed names

fitted change in bp

the model's one-step value

the last value carried forward

Never called

a forecast or a prediction

a projection or an outlook

a signal, a call, a view on rates

Every sentence you write is in the past tense: what a model did on dates inside the file.

The order of the work



five walk-forward folds

tested, committed

opened once

The notebook enforces it

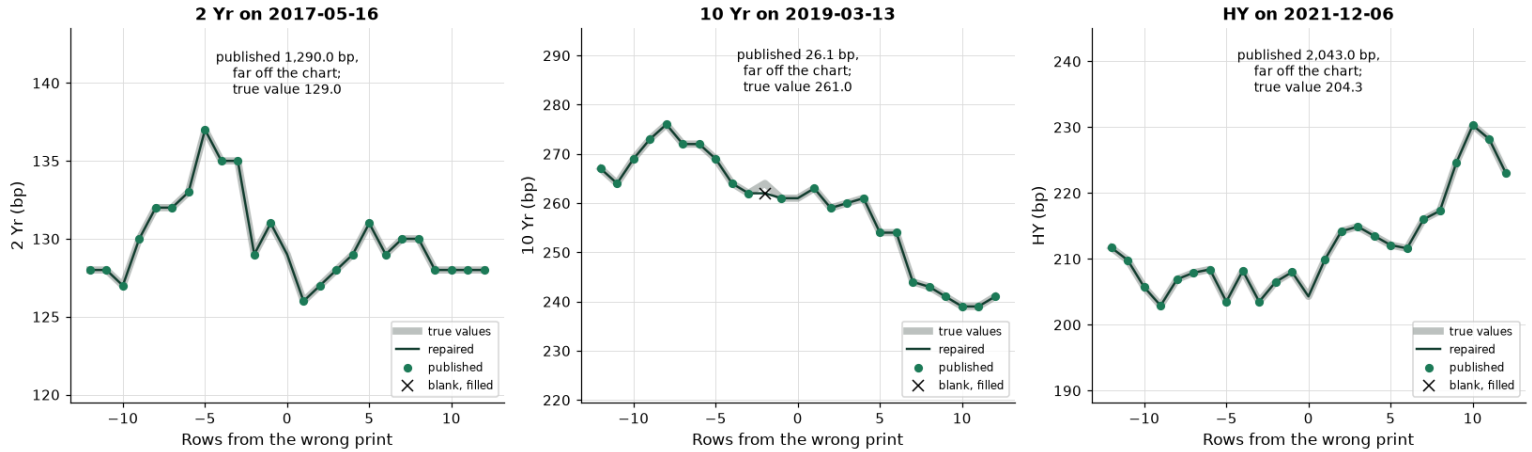
open_held_out_rows() stops before its first use unless choice.md is committed, and stops on a second use.

Why

Opening the years twice makes them more development years. If the choice cannot be shown to come before the score, the score is not a test of the choice.

Folds that look back, features built from earlier rows, and now a choice with a date on it.

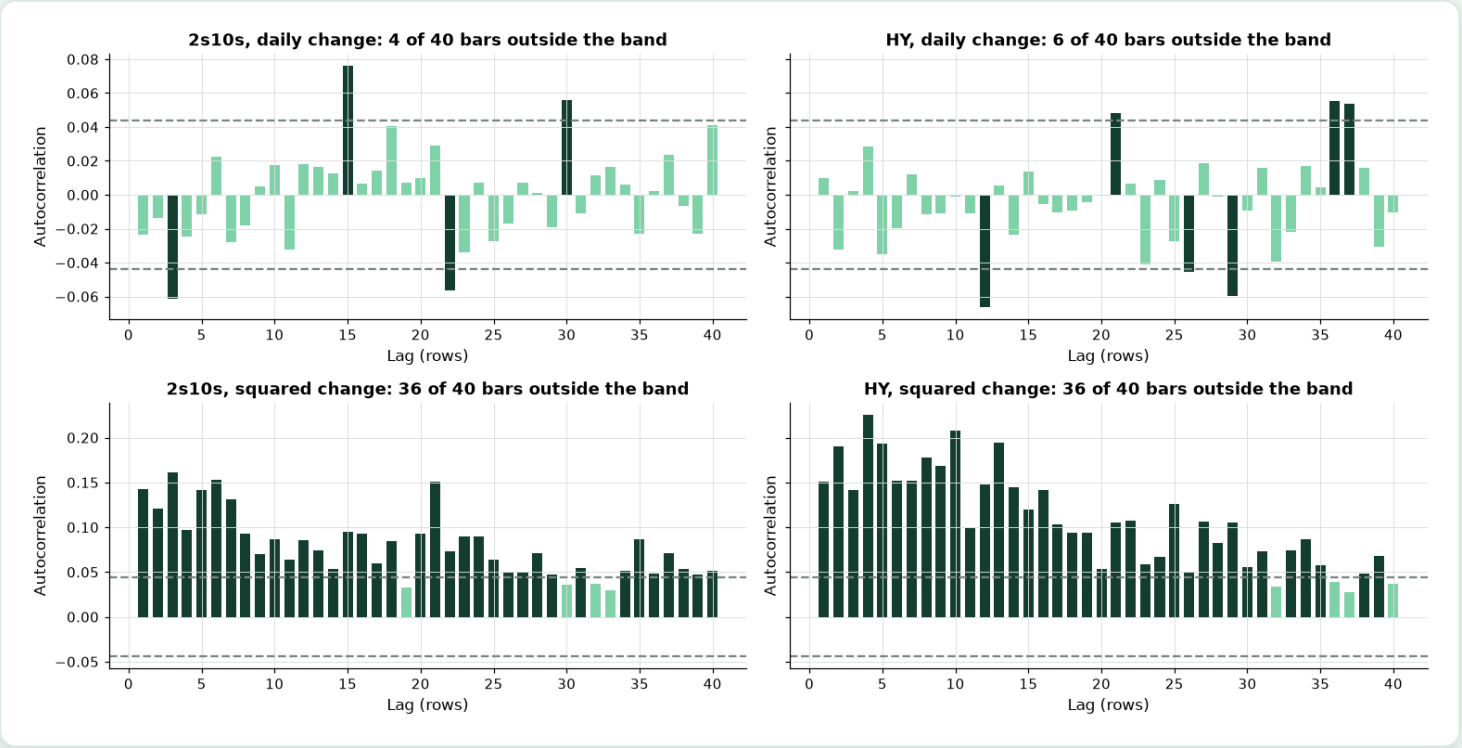
The repair



date	column	old	new
2017-05-16	2 Yr	1,290.0	129.0
2015-09-16	2 Yr	blank	82.0
2019-03-13	10 Yr	26.1	261.0
2015-10-06	10 Yr	blank	205.0
2021-12-06	HY	2,043.0	204.3
2016-01-20	HY	blank	217.5

12 blank cells and three wrong prints. A second feed corrected six cells and wrote the log; a forward fill of one row repaired the other nine.

Stationarity and memory

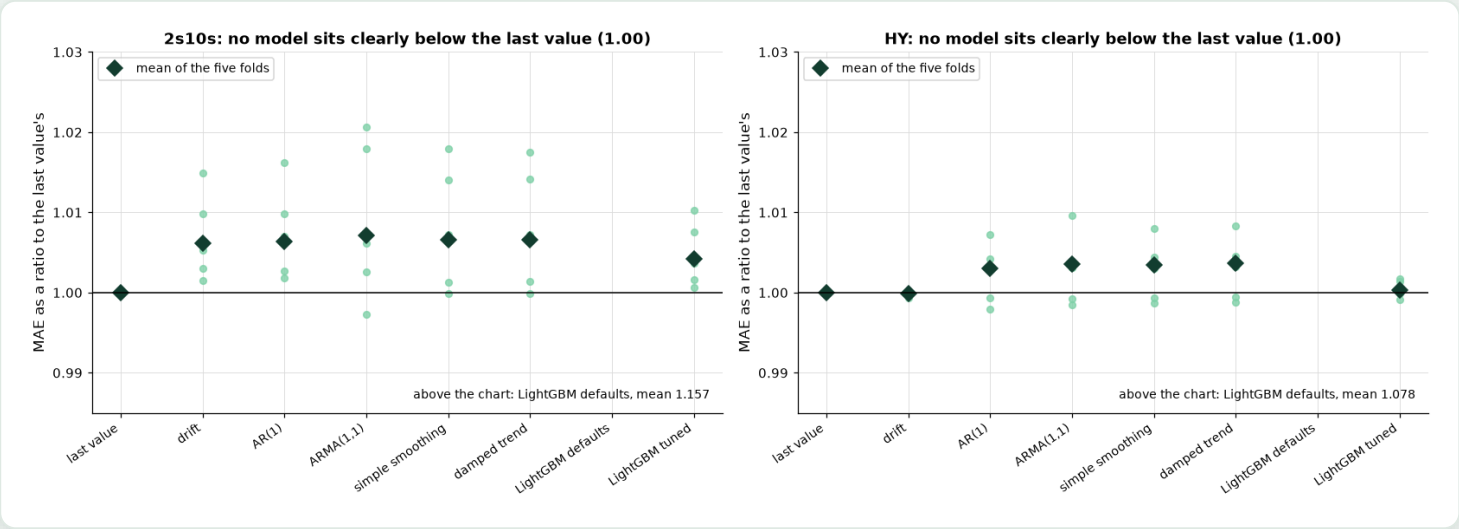


Series	Form	ADF p	KPSS p
2s10s	levels	0.7900	0.01
2s10s	changes	0.0000	0.10
HY	levels	0.0791	0.01
HY	changes	0.0000	0.10

Changes: 4 and 6 of 40 bars outside the band. Squared changes: 36 and 36 of 40. The level of a change has almost no memory; its size does.

Both tests read the levels as a unit root and the changes as stationary, so the models are fitted on the changes.

The baselines and the models of the level

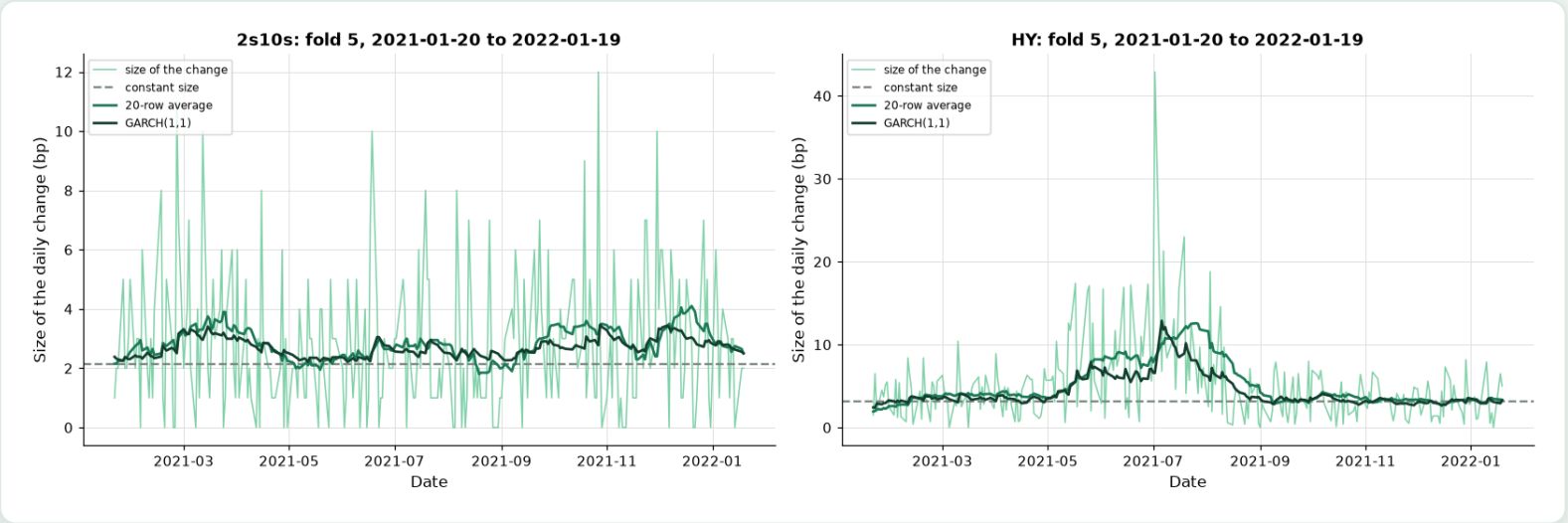


Model	2s10s	MASE	HY	MASE
last value	2.1595	1.0000	3.5002	1.0000
drift	2.1728	1.0062	3.4999	0.9999
ARMA(1,1)	2.1750	1.0071	3.5126	1.0035
simple smoothing	2.1738	1.0066	3.5124	1.0035
LightGBM defaults	2.4994	1.1574	3.7746	1.0784
LightGBM tuned	2.1686	1.0042	3.5014	1.0003

Five folds, 504 dates to start and 252 to score at a time. No model for the 2s10s was closer than the last value on the mean; for the HY only the drift was, by 0.0003 bp and in 2 folds of 5.

Mean MAE in bp, then MASE (1.0 is the last value). LightGBM with its defaults is above the chart.

Volatility: the size of a change



Ratio to the constant	2s10s	HY
20-row average	0.9724	0.9736
GARCH(1,1)	0.9747	0.9534
GARCH folds below (of 5)	3	3

GARCH(1,1) persistence 0.9940 (2s10s) and 0.9660 (HY). The size of a change had something to find: a few percent of gain, not a landslide.

Development folds only. The held-out years are still closed.

How the choice rule works

The input	A table: one row per fold, one column per model, and the baseline's name
The rule	A model replaces the baseline only if its mean is lower AND it is lower in at least 3 of 5 folds
The winner	Among those that qualify, the lowest mean; with none, the baseline stays
The refusals	A baseline that is not a column, min_folds out of range, a blank
What comes out	A column name. A held-out score never goes in

The choice, made on the development years

Series	Level of the change	Size of the change
2s10s	last value	20-row average
HY	last value	GARCH(1,1)

The closest rival, for the HY

The drift: mean 3.4999 bp against 3.5002 for the last value, but lower in only 2 of 5 folds. It did not clear the rule.

On the record

The choice goes in choice.md, with its reasons from the development folds only, and into a commit. That commit lets the held-out years open.

The rule was written and tested before any model was scored.

The held-out years, opened once

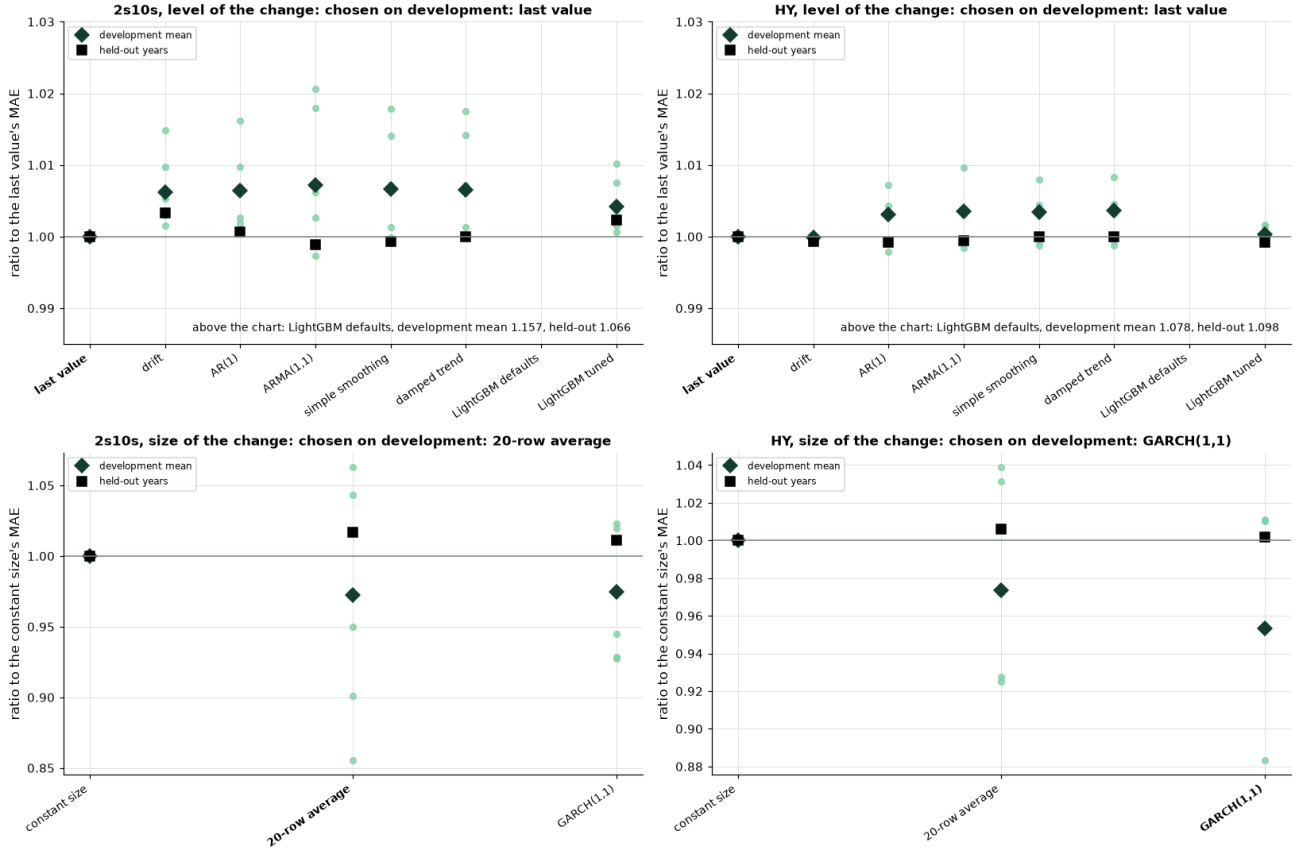
Model	2s10s MAE	gap	HY MAE	gap
last value	2.9542	0.0000	3.9820	0.0000
drift	2.9638	0.0096	3.9794	-0.0026
AR(1)	2.9560	0.0018	3.9789	-0.0031
ARMA(1,1)	2.9507	-0.0035	3.9797	-0.0023
simple smoothing	2.9523	-0.0019	3.9820	0.0000
damped trend	2.9541	-0.0001	3.9820	0.0000
LightGBM defaults	3.1478	0.1936	4.3720	0.3900
LightGBM tuned	2.9609	0.0067	3.9787	-0.0033

939 dates, 3 Jan 2023 to 2 Oct 2026

Each model fitted once on all the development dates, then run through the held-out dates with its coefficients held fixed. The last value, the model the folds chose, sits at the top. Gaps are in bp against it: in the third decimal, except LightGBM with its defaults.

A model that scored 0.003 bp lower was not picked by the rule; picking it now would be the leak.

The closing picture



How to read it

Dots: the five development folds. Diamond: their mean. Square: the held-out years, opened once. Top row, the level of the change: the squares sit near the diamonds, at about 1, inside the dots. Bottom row, its size: the development diamonds sit below 1 and the squares above it.

What the models show, and where they fail

What they show

Nothing fitted on past changes of the 2s10s or the HY scored meaningfully closer to the held-out values than the last value carried forward. The size of a change had a few percent of gain on the development folds, and the gain was gone on the held-out years.

Where they fail

Coefficients held fixed over the held-out dates

One stretch of dates, two series

One of them invented

No other models, windows, or lags

Not a view on rates or spreads

The description is written in the past tense, and the check cell stops on words that look forward.

Excel side by side

Excel side by side (tested in Excel)

`=IF(A2="",B1,A2)`	forward fill; 4 blanks counted
`=CORREL(D4:D2002,D3:D2001)`	lag-1 autocorrelation -0.0238
`=SLOPE(D4:D2002,D3:D2001)`	AR(1) slope -0.0237
`=SUMPRODUCT(ABS(D2003:D2941))/COUNT(D2003:D2941)`	last value MAE 2.9542 bp
`=SUMPRODUCT(ABS(D2003:D2941-(K3+K2*D2002:D2940)))/COUNT(D2003:D2941)`	AR(1) MAE 2.9556 bp
`=AVERAGE(F2:F6)/AVERAGE(B2:B6)`	ARMA(1,1) MASE 1.0071

2s10s results; every value equals Python's. Excel has no ARIMA, GARCH, or boosted trees.

Save the day: the order in the history

1. Module

tskit and its tests, passing

2. Rule

choose_model, committed

3. Choice

choice.md, before any held-out row

4. Results

then the results and the pages

```
Deliverable 11: the AI-use log
Deliverable 10: what the models show, and where they fail
Deliverable 10: the models against the last value
Deliverable 8: the choice, made on the development years
Deliverable 8: choose_model and its test
Deliverable 2: the change log of the repair
Deliverable 1: the module, every test passing
```

The choice was committed before the results: Git's order says it was made first.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which dates fit, which dates score, what may be filled: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the ratio divides by the smallest MAE in the table: a last value that is not the best model scores 1.0526 instead of exactly 1.0.

Code that runs is not the same as code that is right.

What you can do now

- Repair with a change log, and fill by a rule that never looks ahead
- Stationarity and memory: the changes have almost none, their squares have a lot
- Baselines, ARIMA, smoothing, LightGBM, and GARCH, walk-forward in bp
- A tested choice rule, committed before the held-out years open
- The held-out years, opened once: no model clearly beat the last value

Course 6 Neural Networks